The elimination method

1

a 7 **b** $x = -3$ **c** $y = \frac{3}{7}$

2

a $x = -\frac{16}{3}, y = 3$ **b** $x = 2, y = -9$

c $x = 4, y = -9$ **d** $x = 4, y = -\frac{13}{3}$

e $x = -4, y = \frac{23}{3}$ **f** $x = 0, y = 3$

g $x = -4, y = -\frac{13}{9}$ **h** $x = 2, y = \frac{11}{7}$

i $x = -2, y = -2$ **j** $x = -20, y = \frac{53}{3}$

k $x = 3, y = -5$ **l** $x = -5, y = 7$

m $x = -\frac{2}{3}, y = \frac{1}{3}$ **n** $x = -\frac{32}{7}, y = -3$

o $x = -\frac{7}{9}, y = -2$ **p** $x = 1, y = -3$

q $x = 5, y = -2$ **r** $x = 3, y = 1$

3

a

i $3x + 11y = -48$ and $-6x + 21y = -76$

ii $x = -\frac{4}{3}$ and $y = -4$

b

i $2x + 3y = 5$ and $5x + 4y = 2$

ii $x = -2$ and $y = 3$

c

i $4x + 3y = 20$ and $8x - 3y = 4$

ii $x = 2, y = 4$

d

i $6x + 5y = 240$ and $x + 3y = 66$

ii $x = 30, y = 12$

4

a $x = 2, y = 8$ **b** $x = 2, y = -9$ **c** $x = 8, y = 3$ **d** $x = 5, y = 5$

e $x = -9, y = 4$ **f** $x = 6, y = 7$ **g** $x = -20, y = 15$ **h** $x = 3, y = \frac{23}{3}$

i $x = -1, y = -1$ **j** $x = \frac{1}{2}, y = \frac{3}{2}$ **k** $p = -\frac{3}{2}, q = \frac{23}{10}$ **l** $x = -2, y = 3$

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Applications

5

a $x + y = 122$ **b** $x - y = 12$ **c** $x = 67$ **d** $y = 55$

6

a $9x + 2y = 26$ **b** $3x + 4y = 22$ **c** $x = \$2$ **d** $y = \$4$

7

a $x = 1$ **b** $y = 68$ **c** 2016 **d** 68%

8

a $2x + y = 27, 5x - y = 43$ **b** $x = 10, y = 7$

9

a $x + y = 10, 6x - 5y = 5$ **b** $x = 5, y = 5$

10

a $7x + y = 64, 3x - y = 16$ **b** $x = 8, y = 8$

11

a $y = 10x, y = 7x + 18$ **b** Son is 6 and mother is 60 years old.

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